Communicating with Data – Physical Visualizations & Visualization Accessibility

Dr. Ab Mosca (they/them)


CS LIAISONS PRESENT
APPLY TO BECOME A CS DEPARTMENT LIAISON!
CONNECT WITH AND HELP OTHER CS STUDENTS
APPLY BY NOV 15TH !

Plan for Today

- Intro to physical visualization
  - Examples
  - Benefits
- Visualization accessibility
  - For various groups
  - Assessing accessibility

Brainstorm

- Have you ever encountered a physical visualization? What was it?
- What might be the benefits of a physical visualization?

Example

2024 – Monarch Butterfly Migration



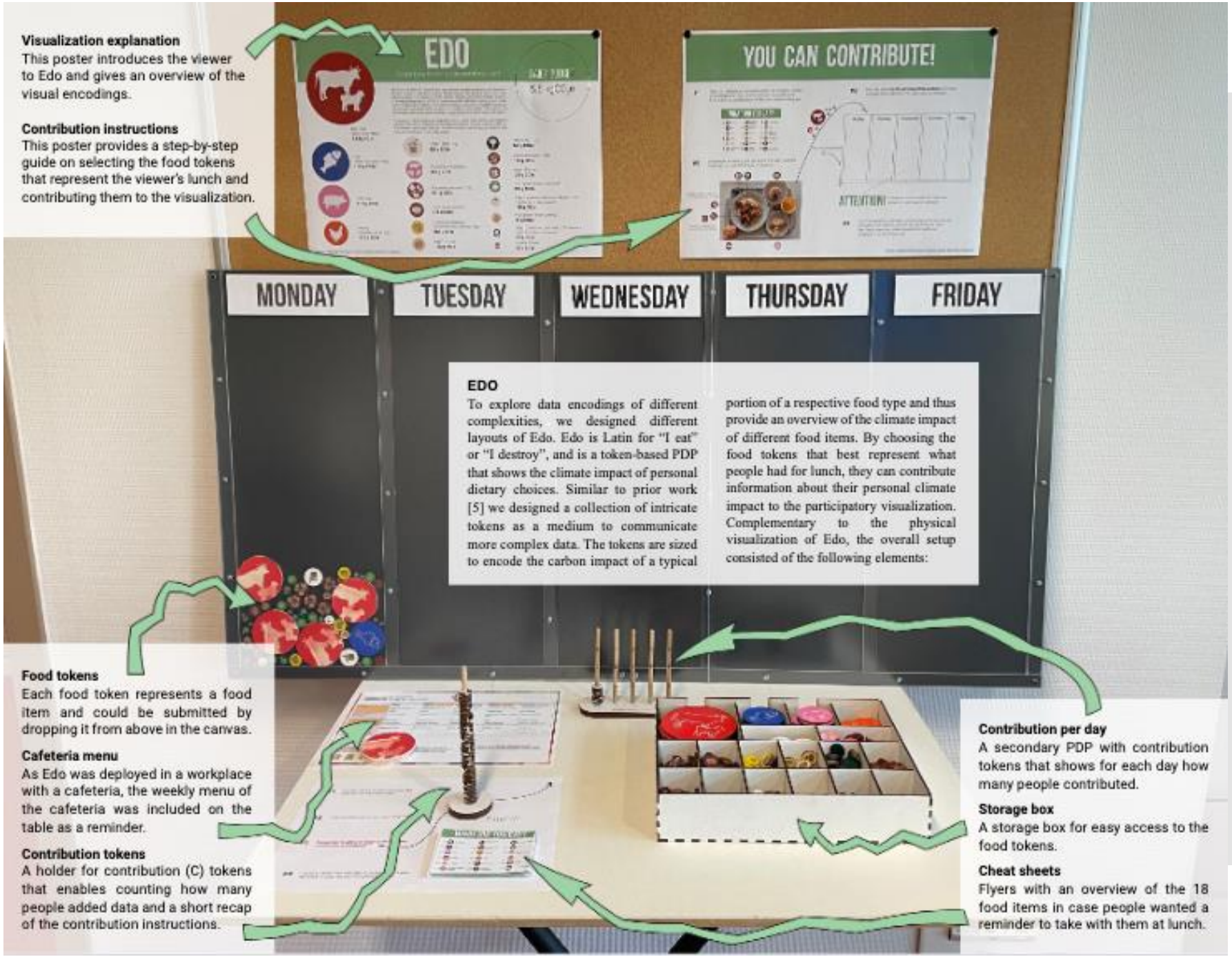
https://www.instagram.com/p/C8ckPxkP8VK/

Example



https://datavisualizationstudio.com/climaticbracelet/en

Example



https://vimeo.com/818729202?fl=pl&fe=sh

Example



https://datavandals.com/

https://youtube.com/shorts/25mtFbsQsbU?si=HbXA46ZjfkPZY5dG

Physical Visualization

A data physicalization is *a physical artifact whose geometry or material properties encode data*

Benefits:

- Exploit active perception skills (perception related to movement)
- Leverage spatial perception skills
- All senses can participate in information gathering
- Increased accessibility for visually-impaired people
- A large body of evidence from educational psychology suggests manipulable physical representations facilitate understanding and learning
- Anecdotal evidence suggests physicalizations may aid individuals in engaging with and communicating information to others more effectively than with digital representations

Yvonne Jansen, et al.. 2015. Opportunities and Challenges for Data Physicalization. In Proceedings of the 33rd Annual ACM CHI '15. Association for Computing Machinery, New York, NY, USA, 3227–3236. https://doi.org/10.1145/2702123.2702180

Accessible Visualization

Accessible Visualization

- Efforts to augment accessibility in data representations have been predominately directed towards individuals from blind or low-vision (BLV) communities

FREQUENCY OF CODES IN *TARGET AUDIENCE* FOR THE 152-PAPER CORPUS.

Target Audiences	Papers w/ Code	This Code Only
BLV	119 (78.3%)	118 (77.6%)
Motor/Physical	12 (7.9%)	8 (5.3%)
Older Adults	6 (4.0%)	3 (2.0%)
DHH	5 (3.3%)	5 (3.3%)
General Disability	5 (3.3%)	5 (3.3%)
Autism	2 (1.3%)	1 (0.7%)
Cognitive	2 (1.3%)	1 (0.7%)
IDD	2 (1.3%)	0 (0.0%)
Learning Disabilities	2 (1.3%)	2 (1.3%)
Other	2 (1.3%)	2 (1.3%)
ADHD	1 (0.7%)	1 (0.7%)

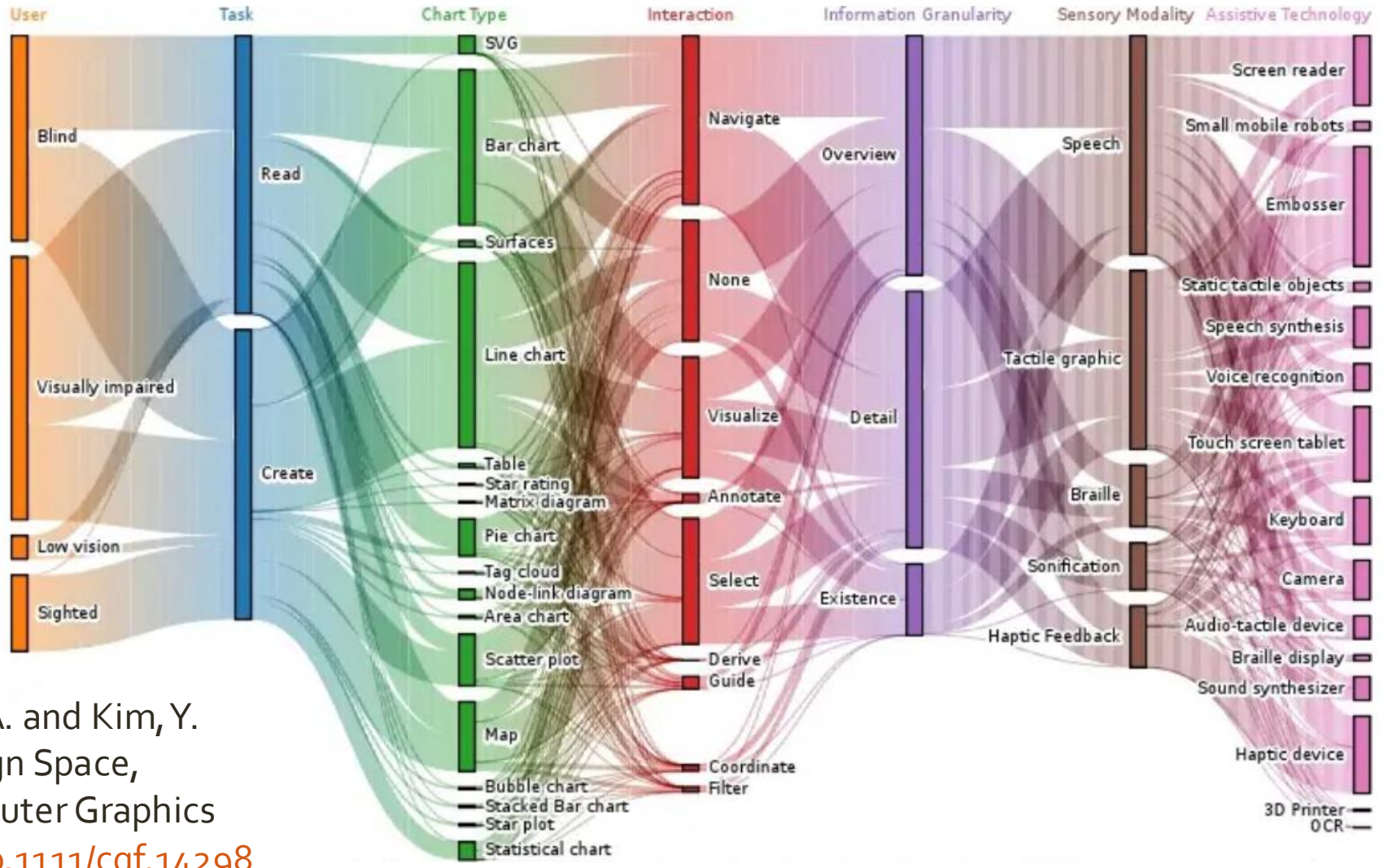
'This code only' counts instances where the code was applied in isolation, not alongside any other codes in this dimension.

B. L. Wimer, et al., "Beyond Vision Impairments: Redefining the Scope of Accessible Data Representations," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 12, pp. 7619-7636, Dec. 2024, doi: 10.1109/TVCG.2024.3356566

Accessible Visualization

BLV Accessibility

- Design space



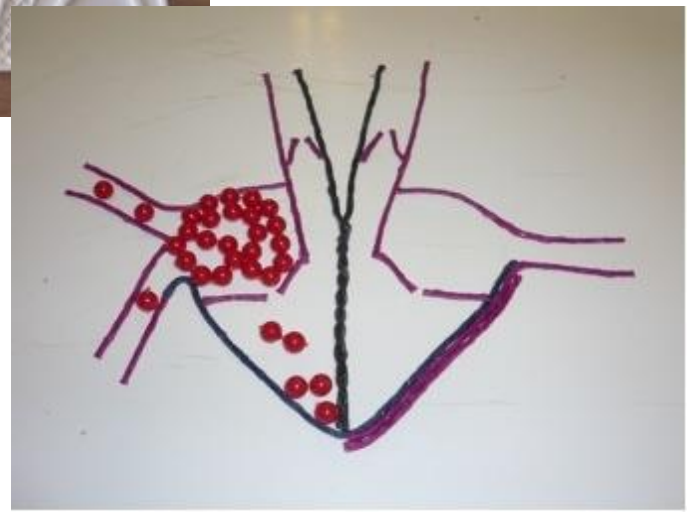
Kim, N.W., Joyner, S.C., Riegelhuth, A. and Kim, Y. (2021), Accessible Visualization: Design Space, Opportunities, and Challenges. Computer Graphics Forum, 40: 173-188. https://doi.org/10.1111/cgf.14298

Accessible Visualization

BLV Accessibility

- Convert visual encoding into auditory or tactile encoding







https://www.perkins.org/resource/tactile-models-tactile-graphics/

Accessible Visualization

BLV Accessibility

- Enhance visualization compatibility with screen readers
  - Ex. Chart Reader

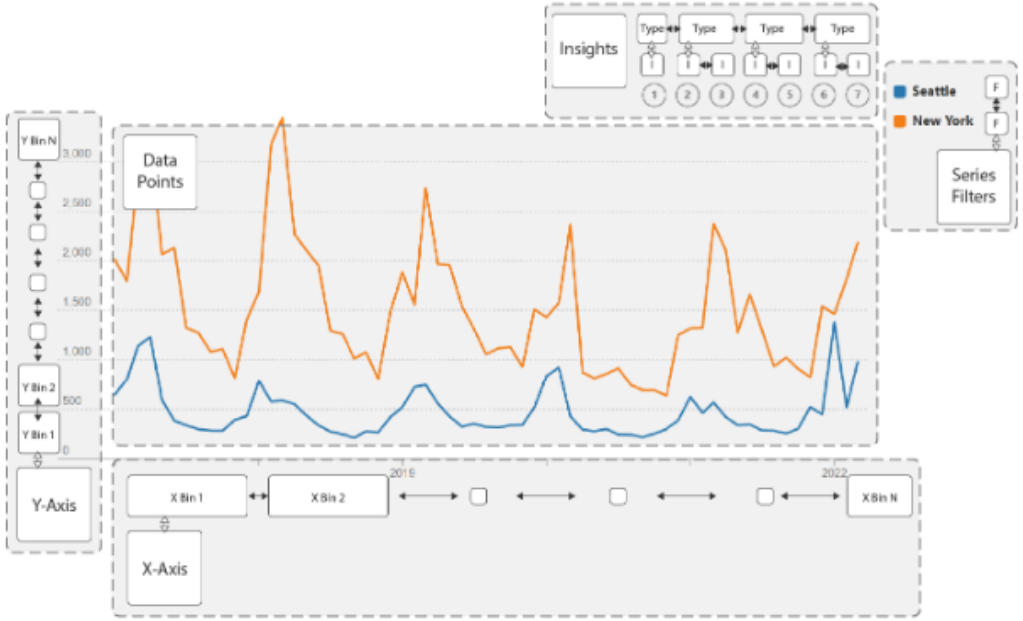


Figure 2: Five regions in charts rendered using Chart Reader to support accessible visualization experience. Data Insights, with sub-regions for each insight type and insight, further subdivided for each individual insight. X-Axis, with sub-regions for each bin along the axis. Y-Axis, with sub-regions for each bin along the axis. Data Points region. Filters, with sub-regions for each individual series.

John R Thompson, et al.. 2023. Chart Reader: Accessible Visualization Experiences Designed with Screen Reader Users. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 802, 1–18. https://doi.org/10.1145/3544548.3581186

Accessible Visualization

Motor Disability Accessibility

- Gaze-based input



(a) Gaze Fisheye : when a viewer's gaze falls on the metro map, the stations and their names around the point of gaze will be magnified

Y. Xie, et al., "GazeMetro: A gaze-based interactive system for metro map," in Proc. 23rd Int. ACM SIGACCESS Conf. Comput. Accessibility, New York, NY, USA: Association for Computing Machinery, 2021, doi: 10.1145/3441852.3476569.

Accessible Visualization

Motor Disability Accessibility

- Voice-activated conversational user interfaces



Yichen Han,et al. 2022. Towards Visualization of Time–Series Ecological Momentary Assessment (EMA) Data on Standalone Voice–First Virtual Assistants. In Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22). Association for Computing Machinery, New York, NY, USA, Article 60, 1–4. https://doi.org/10.1145/3517428.3550398

Accessible Visualization

Neurodiverse Accessibility

- Very little research on ADHD accessibility in visualization
  - So far, no solid design guidelines

B. L. Wimer, et al., "Beyond Vision Impairments: Redefining the Scope of Accessible Data Representations," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 12, pp. 7619-7636, Dec. 2024, doi: 10.1109/TVCG.2024.3356566

Accessible Visualization

Neurodiverse Accessibility

- Little, but some research on accessibility to autistic users

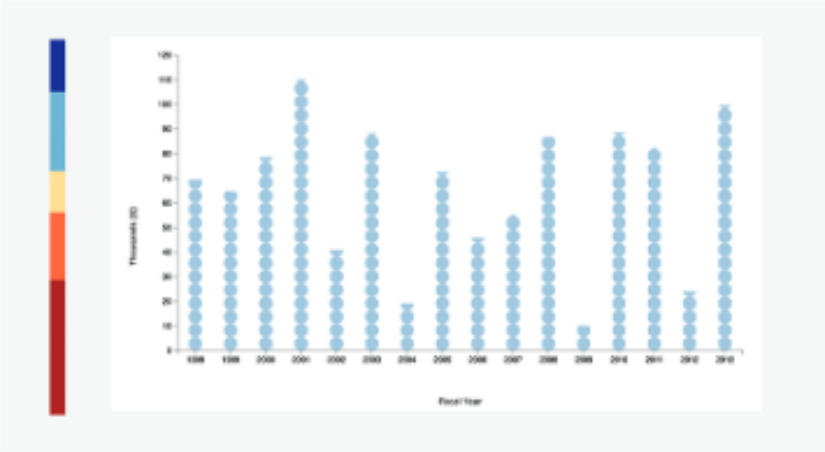
Main Theme	Challenges	Impact
Visual Elements Design	**Cluttered Visuals & Layout:** Overly busy interfaces, ads, side panels, or dense arrangements	Increased cognitive load, difficulty maintaining focus, frustration, disengagement, difficulty identifying essential information, disorientation.
	Ambiguity in Visualisation Design: Incomplete shapes, inconsistent formats or layouts, unfamiliar graph types	Confusion, discomfort, reduced clarity, difficulty trusting/interpreting data, lower engagement.
	Colour and Contrast Issues: Bright colours, low contrast, white backgrounds	Visual fatigue, eye strain, increased cognitive effort, sensory discomfort, slowed reading speed, avoidance of visualisation tasks.
	Proportional Representation Using Shapes: Shape size to represent data	Difficulty interpreting proportions, reduced clarity, cognitive overload, mistrust of accuracy, disregard for shape-based data.
	Font Colour and Size: Low-contrast text and small fonts	Reduced readability, increased effort, loss of focus, misinterpretation of content.
Dynamic Elements	**Animations and Transitions:** Transitions, pop-ups, auto-scrolling content	Distraction, cognitive load, visual fatigue, frustration, discomfort, lower comprehension, disengagement.
Navigation and Information	**Lack of Essential Context:** Missing labels, undefined symbols, absent legends	Confusion, misinterpretation, frustration, uncertainty, increased effort.
	Navigation Challenges: Poorly visible controls, unintuitive menus	Interaction difficulties, inefficiency, loss of control, irritation, usability issues.
Cognitive and Perceptual Differences	**Co-occurring Conditions:** Unaccommodated needs (e.g. ADHD, coordination disorders)	Difficulty with sliders, large numbers, horizontal layouts; increased cognitive demand.
	Details Focused: Absence of explicit key takeaways, hidden critical information (trends, comparisons) not immediately visible	Confusion, high effort to extract meaning, frustration.

M. Alzahrani, et al., "Exploring Data Visualisation Barriers and Needs of Autistic Adults," *2025 IEEE Workshop on Accessible Data Visualization (AccessViz)*, Vienna, Austria, 2025, pp. 10-13, doi: 10.1109/AccessViz68666.2025.00007

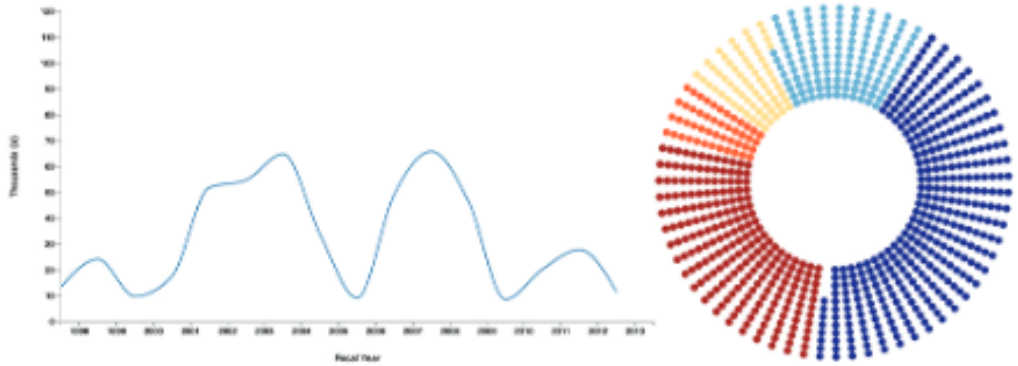
Accessible Visualization

IDD Accessibility

- Avoid pie charts, use familiar metaphors, manage visual complexity, use discrete encodings for axis-aligned representations



Accessible



Not Accessible

K. Wu, et al.. Understanding Data Accessibility for People with Intellectual and Developmental Disabilities. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, pp. 1–16. ACM, Yokohama Japan, May 2021. doi: 10.1145/3411764.3445743 1

Evaluating Accessibility

Chartability

- An accessibility evaluation system specific to data visualizations and interfaces which aims to help practitioners answer the question, "how accessible is my data visualization?"
- Organizes knowledge from disparate bodies of work into testable heuristics based on the functional accessibility principles POUR (Perceivable, Operable, Understandable, and Robust) and 3 novel principles CAF (Compromising, Assistive, and Flexible)

Elavsky, Frank & Bennett, Cynthia & Moritz, Dominik. (2022). How accessible is my visualization? Evaluating visualization accessibility with Chartability. Computer Graphics Forum. 41. 57-70. 10.1111/cgf.14522.

Evaluating Accessibility

Chartability

- Critical heuristics

Heuristic Title	Principle
Low contrast	Perceivable
Small text	Perceivable
Content is only visual	Perceivable
Interaction has only one input	Operable
No interaction cues/instructions	Operable
No explanation for how to read	Understandable
No title, summary, or caption	Understandable
No table	Compromising
Data density inappropriate	Assistive
User style change not respected	Flexible
… +35 non-Critical heuristics	

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Evaluating Accessibility

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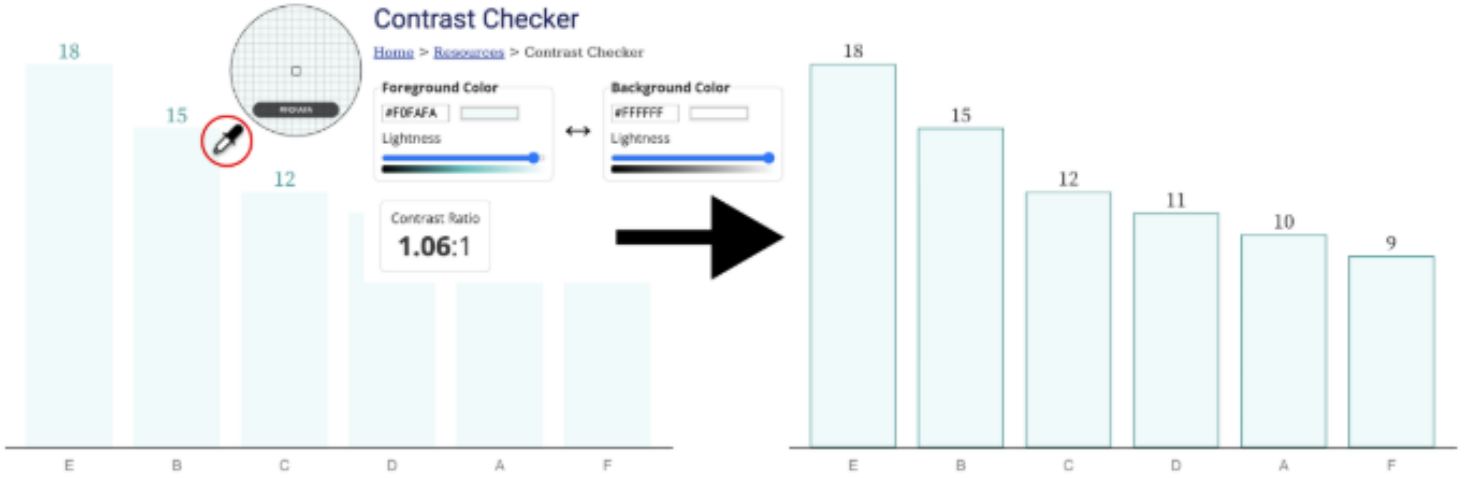


Figure 1: A low contrast chart (left) compared to a higher contrast version (right). A dropper tool is extracting the fill color of the bar and then a contrast ratio has been calculated. Note that the fill color is the same on both bars, but darker borders have been added to ensure the visualization passes contrast tests.

Elavsky, Frank & Bennett, Cynthia & Moritz, Dominik. (2022). How accessible is my visualization? Evaluating visualization accessibility with Chartability. Computer Graphics Forum. 41. 57-70. 10.1111/cgf.14522.

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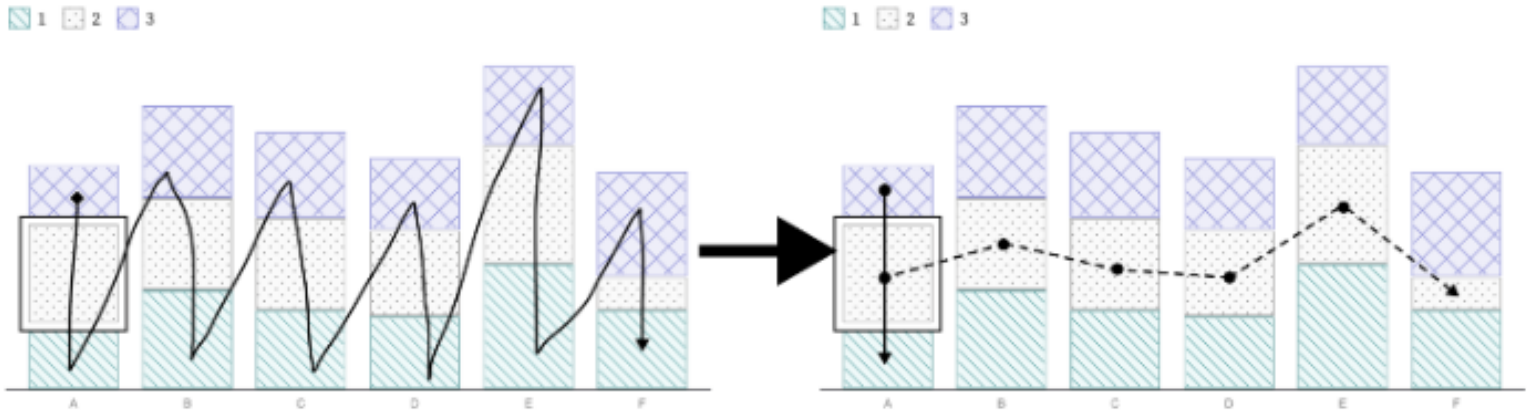


Figure 2: Keyboard navigation paths on a stacked bar chart. The left shows a serial navigation example, typically just a default of rendering order. The right shows both groups (the stack of bars) and categories (the color/texture shared among bars across stacks) as dimensions to explore laterally or vertically.

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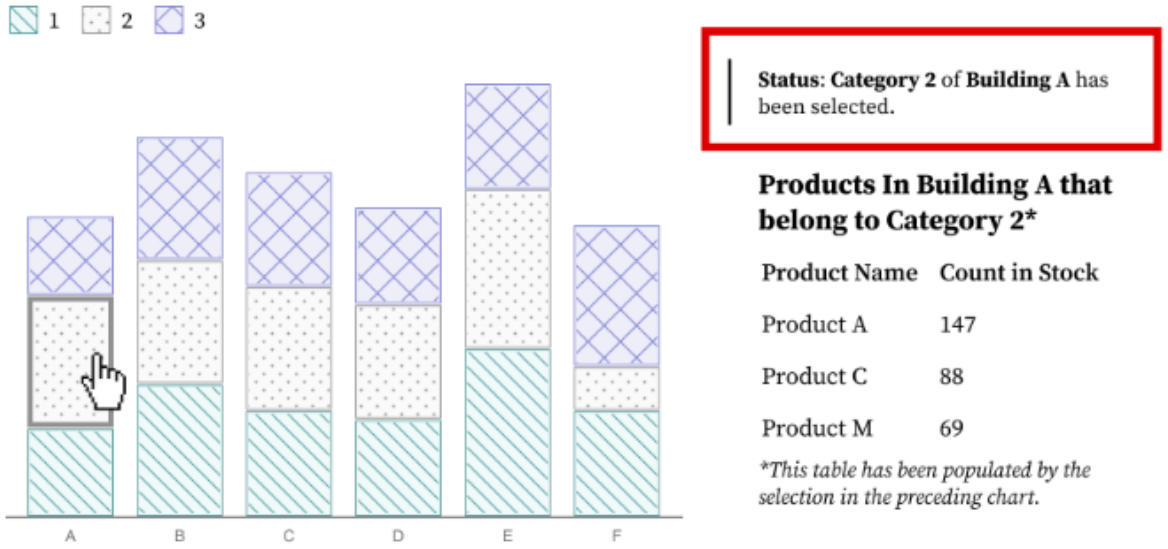


Figure 3: A mouse cursor is selecting a bar (left, shown with a thick indication border) in a stacked bar chart to filter a dataset (on the right). A system alert (red box) notifies the user of their interaction result. This selection capability must also be provided for the keyboard interface and the alert must be announced to screen readers.

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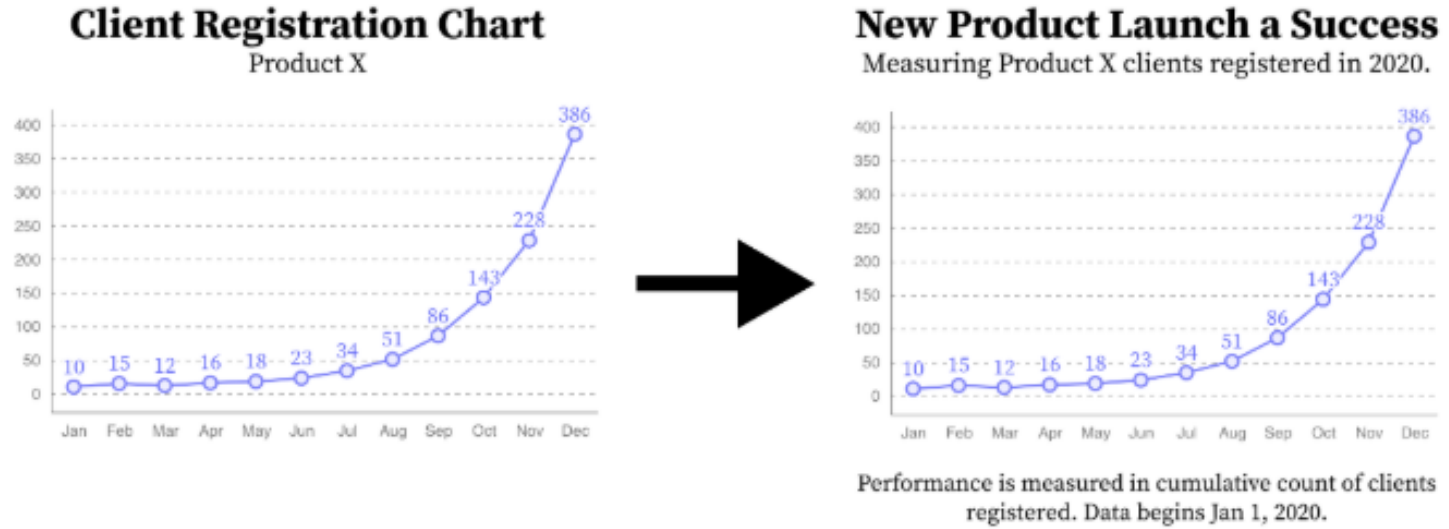


Figure 4: Charts must have a visually available textual explanation provided that summarizes the outcome. “Client Registration Chart” for “Product X” (left) is inaccessible while “New Product Launch a Success” (right) gives a clear takeaway.

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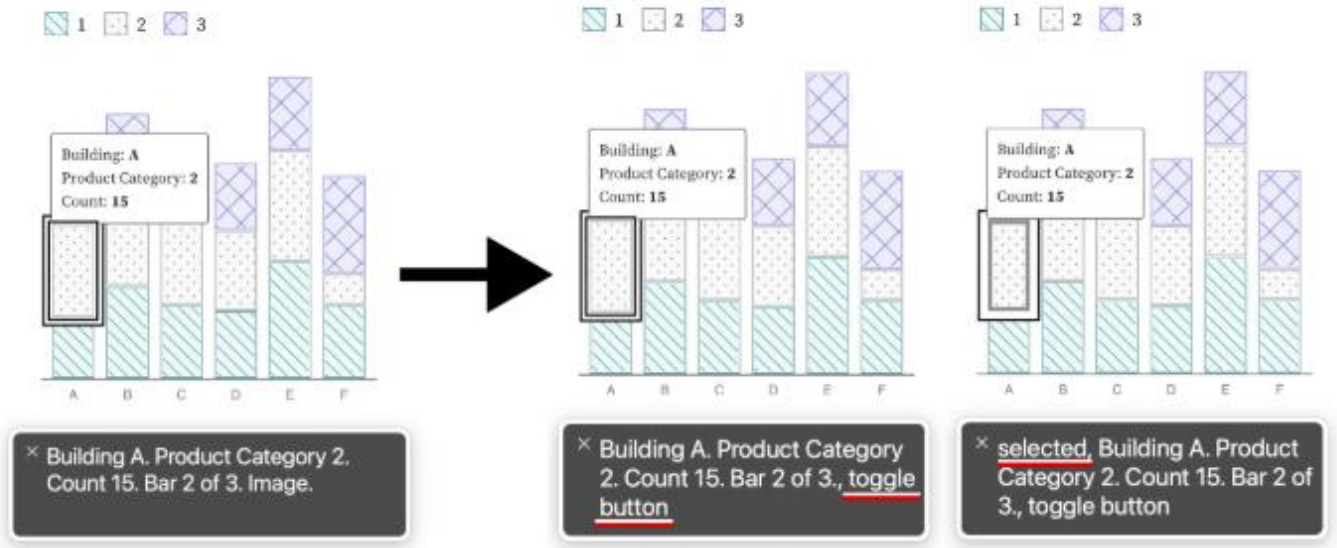


Figure 5: An interactive chart displaying only "Image" as semantic information with no feedback provided on selection. The robust semantics given to a screen reader, "toggle button" (middle) as well instant feedback, "selected" (right) are considered proper semantics for an interactive experience.

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Evaluating Accessibility

Chartability



Month	Registered Clients
Jan	10
Feb	15
Mar	12
Apr	16
May	18
Jun	23
Jul	34
Aug	51
Sept	86
Oct	143
Nov	228
Dec	386

Figure 6: A line chart (left) with a single line and an accompanying data table (right). This line chart would not provide enough low-level information about each datapoint without the table provided. A table alone however would also be inaccessible. Providing both can satisfy conflicting accessibility needs for different audiences.

Elavsky, Frank & Bennett, Cynthia & Moritz, Dominik. (2022). How accessible is my visualization? Evaluating visualization accessibility with Chartability. Computer Graphics Forum. 41. 57-70. 10.1111/cgf.14522.

Evaluating Accessibility

Chartability

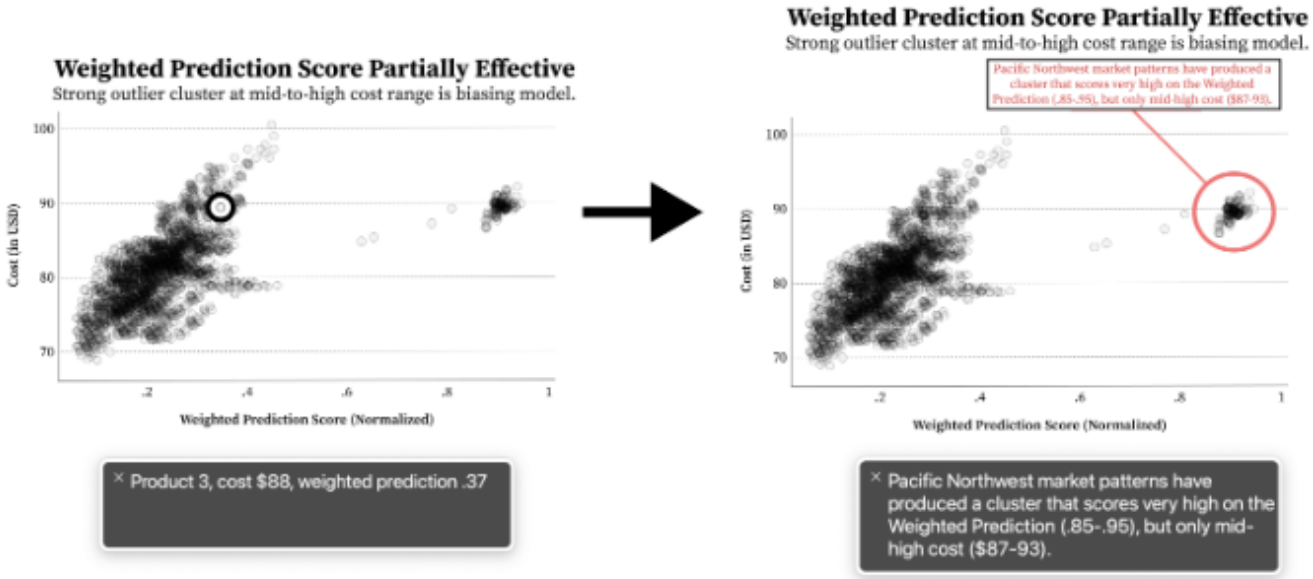


Figure 7: A scatterplot with many points, where a single point within the chart can be accessed by a screen reader (left). Navigating this data piece by piece is unnecessarily tedious, so an annotation callout is provided to help the reader focus on an outlier cluster (right). The callout is being accessed by a screen reader, which is displaying the annotation's summary as well.

Elavsky, Frank & Bennett, Cynthia & Moritz, Dominik. (2022). How accessible is my visualization? Evaluating visualization accessibility with Chartability. Computer Graphics Forum. 41. 57-70. 10.1111/cgf.14522.

Evaluating Accessibility

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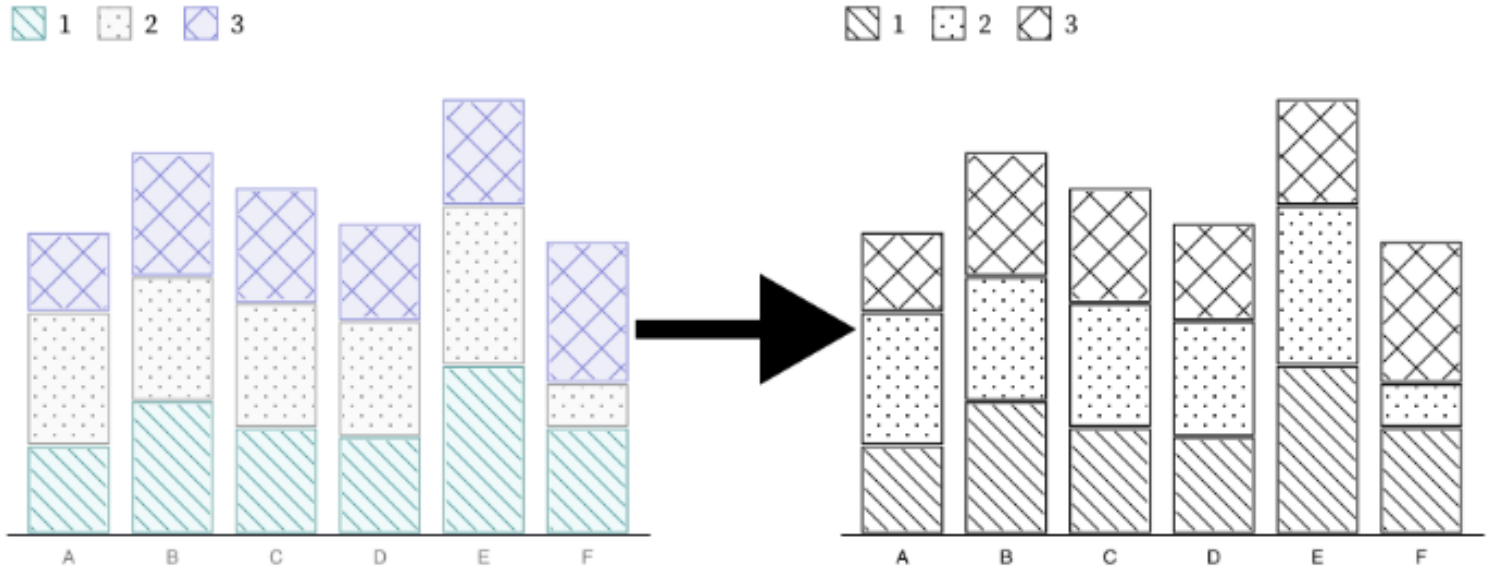


Figure 8: A bar chart with categories (left) shown not conforming to Windows High Contrast White Mode. High contrast mode on Windows requires limiting color palettes, using only black or white for most elements (shown on the right).

Elavsky, Frank & Bennett, Cynthia & Moritz, Dominik. (2022). How accessible is my visualization? Evaluating visualization accessibility with Chartability. Computer Graphics Forum. 41. 57-70. 10.1111/cgf.14522.

Your Turn!

Find a visualization of interest to you.

Evaluate its accessibility via Chartability heuristics.

How would you re-design it to make it more accessible?